



Project Title: Knowledge Graph for E-commerce Customer Behaviour and Financial Analysis

Name: Aman Bhardwaj

Mode: ☐ 6 ECTS / ☒ 3 ECTS

Representations

(LO1) Understand and apply Knowledge Graph Embeddings	☒ I showed basic proficiency ☐ exceeded basic proficiency
I applied Node2Vec to learn 64-dimensional vector representations of entities in the Knowledge Graph. I evaluated the resulting embeddings using a project-defined structural similarity evaluation based on graph connectivity and nearest-neighbour relationships.	Section 3.6
(LO2) Understand and apply logical knowledge in KGs	☐ I showed basic proficiency ☒ exceeded basic proficiency
I implemented four logical forward-chaining rules to derive new customer-product, customer-category, customer-seller, and order-sentiment relationships. The reasoning process uses indexed graph traversal, multiple iterations, and stops at a fixed point when no further facts can be derived.	Section 3.5
(LO3) Understand and apply Graph Neural Networks	☐ I showed basic proficiency ☐ exceeded basic proficiency
Not covered in this project. The project uses Node2Vec for graph embeddings, but does not implement or evaluate a Graph Neural Network.	Not covered.
(LO4) Compare different Knowledge Graph data models from the database, semantic web, machine learning and data science communities.	☒ I showed basic proficiency ☐ exceeded basic proficiency
I compared relational databases, RDF/Semantic Web representations, property graphs, and document databases with respect to representation, relationships, schema flexibility, querying, and graph traversal. I then justified the use of a property graph for this project and discussed its connection to Node2Vec-based graph representation learning.	Section 3.2



Systems

(LO5) Design and implement architectures of a Knowledge Graph	☒ I showed basic proficiency ☐ exceeded basic proficiency
I designed and implemented a layered Knowledge Graph architecture covering data loading and integration, graph construction, enrichment, logical reasoning, graph embeddings, analytics, and service presentation. The resulting graph integrates customers, orders, products, categories, sellers, payments, reviews, sentiment, and geographic information.	Section 3.1
(LO6) Describe and apply scalable reasoning methods in Knowledge Graphs	☒ I showed basic proficiency ☐ exceeded basic proficiency
I implemented indexed forward chaining rather than using unrestricted graph joins, allowing the four reasoning rules to be applied efficiently over the large graph. The reasoning process reached a fixed point after three iterations and added 362,830 inferred relationships; scalability is nevertheless limited by the in-memory NetworkX representation.	Section 3.5
(LO7) Apply a system to create a Knowledge Graph	☐ I showed basic proficiency ☒ exceeded basic proficiency
I implemented an end-to-end pipeline that integrates the Olist datasets and creates a multi-layer Knowledge Graph with 442,401 nodes and 1,085,835 relationships after enrichment and reasoning. The pipeline also validates the graph structure and exports the resulting KG for downstream analysis and services.	Section 3.1
(LO8) Apply a system to evolve a Knowledge Graph	☒ I showed basic proficiency ☐ exceeded basic proficiency
I evolved the initial Knowledge Graph through additional sentiment, geographic, and logical enrichment. In particular, TextBlob-based sentiment information and geographic clusters are added as new knowledge, while logical forward chaining derives additional relationships.	Sections 3.4 and 3.6



Applications

(LO9) Describe and design real-world applications of Knowledge Graphs	☐ I showed basic proficiency ☒ exceeded basic proficiency
I designed a real-world e-commerce Knowledge Graph application for analysing customer behaviour, products, categories, sellers, reviews, transactions, and geographic relationships. The KG supports analytical queries, customer and product analysis, sentiment analysis, graph exploration, and interactive services through a Streamlit application.	Sections 1 and 3.7
(LO10) Describe financial Knowledge Graph applications	☒ I showed basic proficiency ☐ exceeded basic proficiency
I extended the e-commerce KG with financial analysis of payments, customer spending, order values, repeat purchasing, and seller sales. The project demonstrates how financial information can be connected to customer, order, product, and seller entities in a Knowledge Graph, while keeping the analysis descriptive rather than presenting it as a formal financial or predictive model.	Section 3.3
(LO11) Apply a system to provide services through a Knowledge Graph	☐ I showed basic proficiency ☒ exceeded basic proficiency
I implemented a Streamlit-based Knowledge Graph service that provides natural-language-style analytical queries, graph exploration, entity search, analytics, and graph metrics. The query interface uses deterministic intent detection and graph traversal to answer questions directly from the Knowledge Graph.	Section 3.7
(LO12) Describe the connections between Knowledge Graphs (KGs), Machine Learning (ML) and Artificial Intelligence (AI)	☒ I showed basic proficiency ☐ exceeded basic proficiency
I connected Knowledge Graphs with several AI/ML techniques: Node2Vec provides graph-based representations, TextBlob provides sentiment enrichment, logical reasoning derives new knowledge, and analytical services use the resulting KG structure. The project therefore demonstrates how symbolic graph knowledge and data-driven methods can complement each other.	Sections 3.6 and 6



Additional Information

HAS NO EFFECT ON MARKING!

(please fill it out honestly, even if it is less than what is suggested in the ECTS breakdown – you are not judged on time spent!)

How many hours did you spend on your mini-project ?	50 hours
--	----------

* please exclude any hours you spent on parts reused from other courses

How many hours did you spend on your portfolio document preparation (this PDF) ?	15 hours
---	----------

* please exclude any hours you spent on parts reused from other courses

Declaration

I have marked all parts generated by Generative AI (e.g., ChatGPT) or otherwise copied from other sources and given any prompt I used either in a footnote or in an appendix making clear which parts are generated by which prompts or similar.	☒ I confirm this
--	--

Please indicate if you have used generative AI parts of the mini-project	☐ I used it for roughly <XX>% of the mini-project
<If this is the case, please describe here what parts you generated and what you did yourself to fulfil your learning outcomes.>	

Please indicate if you have used generative AI parts of this document	☐ I used it for roughly <XX>% of this document
<If this is the case, please describe here what parts you generated and what you did yourself to fulfil your learning outcomes.>	



Portfolio Report

Knowledge Graph for E-commerce Customer Behaviour and Financial Analysis

1. Introduction

This project develops a Knowledge Graph (KG) for analysing customer behaviour and financial activity in an e-commerce environment using the Brazilian Olist e-commerce dataset. The main goal is to integrate customers, orders, products, categories, sellers, payments, reviews and geographical information into one connected graph that supports analytical queries, logical reasoning and graph-based machine learning.

The project focuses primarily on creating and applying a Knowledge Graph (LO7) and demonstrating a real-world KG application (LO9). Additional work covers Knowledge Graph embeddings (LO1), logical reasoning (LO2), KG data-model comparison (LO4), KG architecture (LO5), reasoning methods (LO6), KG enrichment (LO8), financial KG applications (LO10), KG-based services (LO11), and the relationship between KGs and machine learning (LO12). Graph Neural Networks (LO3) were outside the project scope.

The final pipeline completed successfully and produced a Knowledge Graph containing 442,401 nodes and 1,087,297 relationships. Of these, 723,005 are explicit relationships and 364,292 are inferred or derived relationships, consisting of 362,830 logical relationships and 1,462 embedding-derived relationships.

2. Background and Dataset

E-commerce data contains many interconnected entities. A customer places an order, an order contains products, products belong to categories and are offered by sellers, while orders are connected to payments and reviews. These relationships are difficult to explore through isolated tables, especially when several relationships have to be followed simultaneously. The dataset is directly imported in the code. Also, the public link of the OLIST dataset is: <https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce?resource=download>

A Knowledge Graph provides a natural representation for this type of problem because entities and their relationships can be explicitly represented and traversed. The project therefore uses a property-graph-style representation with NetworkX. The Olist dataset consists of several related CSV files. After validation and cleaning, the main datasets contained:

Dataset	Records
Customers	99,441
Orders	99,441
Order items	112,650
Products	32,951
Payments	103,886
Sellers	3,095
Reviews	99,224



Dataset	Records
Geolocation	738,332
Categories	71

The data-loading stage performs schema validation, duplicate checking, missing-value inspection, type conversion and category translation. Duplicate geolocation records were removed before graph construction.

3. Method

3.1 Knowledge Graph construction (LO7)

The core of the project is the construction of a property graph (LO7). The graph represents entities such as customers, orders, products, categories, sellers, payments, reviews, cities and states.

Important relationships include:

- Customer -[PLACED]-> Order
- Order -[CONTAINS]-> Product
- Product -[BELONGS_TO]-> Category
- Product -[OFFERED_BY]-> Seller
- Order -[HAS_PAYMENT]-> Payment
- Order -[HAS_REVIEW]-> Review
- Review -[HAS_SENTIMENT]-> Sentiment

The initial graph contained 442,390 nodes and 677,982 explicit relationships. Later enrichment added additional entities and relationships.

Concrete construction examples

Three examples illustrate how the source tables were transformed into graph knowledge:

1. **Customer and order construction:** Customer entities were created from the customer identifiers in the customer dataset, while order entities were created from order_id. The order records were then used to create Customer -[PLACED]-> Order relationships.
2. **Product and category construction:** Product entities were created from product_id. Product-category information was used to create Product -[BELONGS_TO]-> Category relationships, with the category translation data providing the English category representation.
3. **Order and payment construction:** Payment records were represented as payment entities and connected to their corresponding orders through Order -[HAS_PAYMENT]-> Payment. This makes payment information directly accessible from the transactional part of the graph.

The overall architecture follows the pipeline:

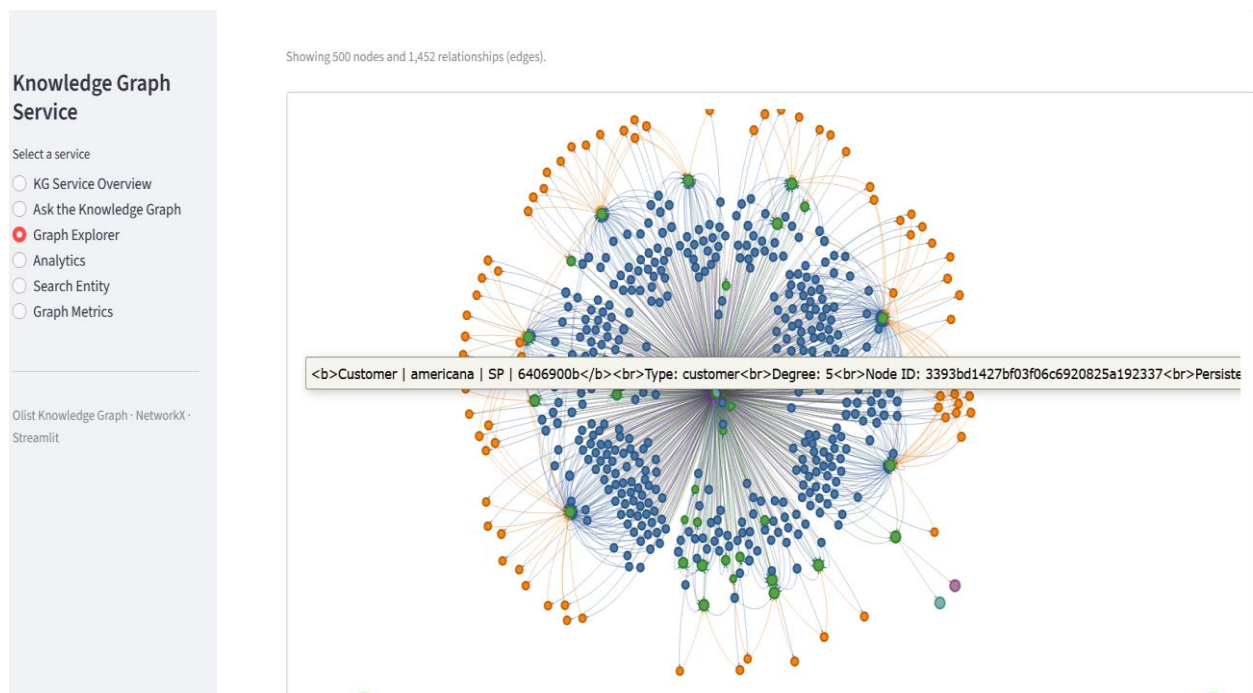
Data loading and cleaning → KG construction → NLP/geographic enrichment → logical reasoning → Node2Vec representation learning → ML-based KG evolution → analytics and query services → Streamlit application.



This layered design separates source-data integration from graph construction, enrichment, reasoning, representation learning and user-facing services (LO5).

After enrichment and logical reasoning, the graph contained **442,401 nodes and 1,085,835 relationships**, including 723,005 explicit relationships and 362,830 logically inferred relationships. The subsequent Node2Vec-based evolution added a further 1,462 embedding-derived relationships, producing the **final KG of 442,401 nodes and 1,087,297 relationships**.

The graph was structurally validated. All nodes had a node-type and all edges had a relationship type. The final graph consisted of one connected component. Below is a visualization of the KG with Depth = 2 and nodes = 500



3.2 Data-model choice (LO4)

The project compares relational databases, RDF/Semantic Web, property graphs and document databases (LO4).

The relational model is useful for the original Olist tables, while RDF provides strong semantic interoperability and ontology-based modelling. Document databases are flexible for document-oriented information. The **property graph** was selected because the project requires direct multi-hop traversal, relationship-oriented analysis and graph-based representation learning.

The chosen graph structure is also used as input to Node2Vec, connecting the data-model decision to the later machine-learning stage.

3.3 Analytical and financial layer (LO9, LO10)

The KG supports customer and business analysis by connecting transaction information with customers, products, categories and sellers.

The final analysis produced:



Measure	Result
Total payment value	16,008,872.12
Average payment value per order	160.99
Orders	99,441
Persistent customers	96,096
Repeat customers	2,997
Products	32,951
Categories	71
Sellers	3,095
Payments	103,886
Reviews	99,224

For example, **2,997 persistent customers** were identified as repeat customers because they were associated with more than one order.

The project also implements a financial KG application (LO10). Payment entities are connected to orders, which indirectly connects financial activity with customers, products, categories and sellers. Customer financial profiles were generated using total payment value, order activity and product/category relationships.

3.4 Knowledge Graph enrichment (LO8)

The KG is extended with derived information, demonstrating KG enrichment and evolution (LO8).

Sentiment enrichment

Review text was analysed using TextBlob. **40,950 review texts** were processed.

The TextBlob results were:

Sentiment	Count
Neutral	37,468
Positive	3,034
Negative	448

The average TextBlob polarity was **0.0272**. The NLP layer adds sentiment properties and HAS_NLP_SENTIMENT relationships while preserving the original review-score sentiment separately. This means the graph can distinguish between sentiment derived from a structured review score and sentiment derived from review text.

Geographic enrichment

The project also analysed **738,332 geographical points** and created **8 derived geographic clusters** with **4,073 city-cluster relationships**.



These clusters are represented in the KG as derived analytical entities and can support regional demand, customer distribution and logistics-oriented analysis.



Embedding-based enrichment

In addition to sentiment, geographic and logical enrichment, the Knowledge Graph also evolved using the learned Node2Vec representations (LO8). A bounded similarity search was applied to 500 candidate product nodes, using a similarity threshold of 0.90, selecting up to three similar products per source node while inspecting at most 50 candidate neighbours. This process added 1,462 EMBEDDING_SIMILAR_TO relationships to the graph.

The graph therefore increased from 1,085,835 relationships after logical reasoning to 1,087,297 relationships after ML-based evolution. These relationships are treated as candidate similarity relationships rather than ground-truth facts, because they are derived from an unsupervised embedding model. This demonstrates how a machine-learning representation can be used to complete and evolve the Knowledge Graph (LO8, LO12).

3.5 Logical reasoning (LO2, LO6)

A central KG component is rule-based logical inference (LO2).

Four rules were implemented. They infer:

1. Customer $\rightarrow$ Product interaction from Customer $\rightarrow$ Order $\rightarrow$ Product.
2. Customer $\rightarrow$ Category association from Customer $\rightarrow$ Product $\rightarrow$ Category.
3. Customer $\rightarrow$ Seller association from Customer $\rightarrow$ Product $\rightarrow$ Seller.
4. Order $\rightarrow$ inferred NLP sentiment from Order $\rightarrow$ Review $\rightarrow$ NLP Sentiment.
5. **Multi-hop customer exploration:** Starting from a customer, the system can follow PLACED $\rightarrow$ CONTAINS $\rightarrow$ BELONGS_TO to identify the product categories associated with that customer's purchases. The same multi-hop exploration can follow PLACED $\rightarrow$ CONTAINS $\rightarrow$ OFFERED_BY to identify associated sellers.



This represents a multi-hop recursive exploration of the graph because information is obtained by following several connected relationships rather than querying a single direct edge.

The reasoning engine uses indexed forward chaining and repeatedly applies the four rules until no new facts can be derived. The process therefore performs multi-hop graph exploration and reaches a fixed point after three iterations. Iteration 1 generated customer-product and order-sentiment relationships, while iteration 2 used the newly available customer-product relationships to derive customer-category and customer-seller relationships. Iteration 3 generated no additional facts, confirming the fixed point (LO2, LO6). The results were:

Reasoning measure	Result
Explicit relationships before reasoning	723,005
New inferred relationships	362,830
Final relationships	1,085,835
Expansion ratio	1.50 times
Relationship increase	50.18%
Reasoning time	17.18 seconds

The process reached a fixed point after three iterations. The third iteration generated no new facts.

This demonstrates how information that is implicit in several graph paths can be converted into explicit inferred knowledge.

3.6 KG embeddings (LO1, LO12)

The final graph was used for graph representation learning with **Node2Vec (LO1)**.

The configuration used:

- 64-dimensional embeddings
- Walk length: 20
- 30 walks per node
- Context window: 10
- $p = 1.0$
- $q = 1.0$

Node2Vec represents nodes as numerical vectors based on their graph neighbourhoods. Similar graph contexts can therefore produce similar vector representations.

For example, the translated/original category representations showed high similarity:

Source node	Similar node	Similarity
CATEGORY_ORIGINAL_cama_mesa_banho	CATEGORY_bed_bath_table	0.9390
CATEGORY_ORIGINAL_beleza_saude	CATEGORY_health_beauty	0.9832



Source node	Similar node	Similarity
STATE_sp	GEOGRAPHIC_CLUSTER_6	0.8978
CATEGORY_bed_bath_table	CATEGORY_ORIGINAL_cama_mesa_banho	0.9390
GEOGRAPHIC_CLUSTER_6	STATE_sp	0.8978

A project-defined structural evaluation using a three-hop criterion produced:

- True positives: **44**
- False positives: **6**
- Structural precision: **0.88**
- Embedding diversity score: **0.7131**

Node2Vec graph embedding technique demonstrates a direct connection between KG structure and machine-learning representation learning (LO12).

3.7 KG-based service and application (LO9, LO11)

The project provides a deterministic Knowledge Graph query service and an interactive Streamlit application (LO11).

The query service supports questions involving:

- customers and orders
- products and categories
- sellers
- payments and average order value
- repeat customers
- customer spending
- sentiment
- customer-specific products, categories and sellers

The Streamlit application provides:

- KG overview
- graph exploration
- entity search
- general analytics
- individual entity analytics
- saved analytical results

The service is deterministic and uses graph traversal and predefined analytical intents rather than an external LLM. It therefore demonstrates how a KG can provide an analytical service directly to users (LO11). A screenshot of a query on KG is as under:



🔍 Ask (Query) the Knowledge Graph

Enter an analytical question in natural language. The deterministic query service identifies a supported intent and answers it using KG traversal and project analytics.

Your question

What orders did customer 0a0a92112bd4c708ca5fde585afaa872 place?

Examples: How many customers are in the KG? · How many orders are there? · What is the total payment value? · Show top 25 repeat customers · Which product categories are most popular? · Which sellers have the most product-order associations? · What is the review sentiment distribution?

Ask

Question answered from the Knowledge Graph.

Answer

This customer has 1 order(s), €13,664.08 in represented payment value, and 1 purchased product(s).

Results

customer_id	customer_unique_id	order_count	product_count	category_count	seller_count	total_spending	average_order_value
1617b1357756262bfa56ab541c47bc16	0a0a92112bd4c708ca5fde585afaa872	1	1	1	1	13664.08	13664.08

🔍 Ask (Query) the Knowledge Graph

Enter an analytical question in natural language. The deterministic query service identifies a supported intent and answers it using KG traversal and project analytics.

Your question

who are top 2 repeat customers?

Examples: How many customers are in the KG? · How many orders are there? · What is the total payment value? · Show top 25 repeat customers · Which product categories are most popular? · Which sellers have the most product-order associations? · What is the review sentiment distribution?

Ask

Question answered from the Knowledge Graph.

Answer

The KG contains 2,997 persistent customers with more than one order. Showing the top 2.

Results

customer_unique_id	customer_record_count	order_count	total_spending	average_order_value
46450c74a0d8c5ca9395da1daac6c120		3	3	9553.02
da122df9eeddfedc1dc1f5349a1a690c		2	2	7571.63

3.8 Execution

The project is designed so that the main Knowledge Graph pipeline can be reproduced from the submitted source code. The Olist Brazilian E-Commerce dataset is downloaded automatically through KaggleHub when main.py is executed.

The main execution sequence is:

1. Create and activate the Python virtual environment.
2. Install the dependencies using requirements.txt.
3. Run python main.py to download the dataset, clean and integrate the data, construct the Knowledge Graph, perform NLP and geographic enrichment, apply logical reasoning, train Node2Vec embeddings, and generate the analytical results.
4. Run python -m streamlit run streamlit_app.py to launch the interactive Knowledge Graph service.

The submitted Readme.md provides the corresponding installation and execution commands. Generated files such as the Knowledge Graph and analytical outputs are produced by the pipeline rather than requiring manual construction.



4. Results

The final pipeline successfully completed all major stages: data validation, graph construction, enrichment, reasoning, validation, Node2Vec embedding, query-service preparation, graph export and visualization. The most important final results are:

Knowledge Graph

- **442,401 nodes**
- **1,087,297 relationships**
- **723,005 explicit relationships**
- **364,292 inferred relationships**
- **1 connected component**
- **Average degree: 4.92**

Customer and financial analysis

- **96,096 persistent customers**
- **2,997 repeat customers**
- **16,008,872.12 total payment value**
- **160.99 average order value**
- **99,441 orders**

KG enrichment

- **40,950 review texts analysed**
- **8 geographic clusters**
- **4,073 geographic relationships**
- **Separate structured and NLP sentiment representations**

Reasoning

- **362,830 inferred facts**
- **50.18% increase in relationships**
- **17.18 seconds reasoning time**
- **Fixed point reached after three iterations**

Embeddings

- **64-dimensional Node2Vec vectors**
- **0.88 project-defined structural precision**
- **0.7131 embedding diversity score**

ML-based KG evolution

- **500 candidate source products**
- **0.90 similarity threshold**



- **Top 3 similar products per source**
- **1,462 EMBEDDING_SIMILAR_TO relationships added**
- **1,085,835 → 1,087,297 relationships**

These results show that the graph is enriched, queried, reasoned over and used as input for representation learning.

5. Learning Outcomes

The strongest emphasis remains on **LO7 and LO9**, as originally planned. **LO2 and LO11** are also strongly demonstrated by the implemented reasoning and application layers. All other LOs demonstrate basic proficiency.

6. Limitations and Discussion

The project has several limitations. First, NetworkX provides an in-memory graph implementation. The project demonstrates a substantial graph with more than one million relationships, but it does not demonstrate distributed or production-scale graph infrastructure.

Second, the reasoning system contains four project-specific rules rather than a complete ontology or comprehensive reasoning system. The reasoning results therefore demonstrate the implemented domain logic rather than general-purpose KG reasoning.

Third, Node2Vec is not a GNN, so LO3 is not addressed. A future extension could investigate Graph Neural Networks and supervised downstream tasks.

Fourth, the embedding evaluation is a project-defined structural proxy. The reported precision of 0.88 should therefore not be interpreted as a standard benchmark result.

Finally, the financial analysis uses the payment information available in the Olist dataset. Payment value is not treated as formal accounting revenue, and customer financial profiles represent historical behaviour rather than predictive financial models.

7. Conclusion

This project successfully developed a Knowledge Graph for e-commerce customer behaviour and financial analysis using the Olist Brazilian e-commerce dataset.

The final system integrates customers, orders, products, categories, sellers, payments, reviews and geographical information into a graph containing 442,401 nodes and 1,087,297 relationships. The project extends the original data through sentiment and geographic enrichment and uses logical rules to derive 362,830 additional relationships.

The implementation demonstrates the creation and application of a Knowledge Graph (LO7, LO9), together with logical reasoning (LO2), bounded forward chaining (LO6), graph embeddings (LO1), data-model comparison (LO4), KG architecture (LO5), KG enrichment (LO8), financial KG analysis (LO10), KG-based services (LO11), and KG/ML integration (LO12).

The main contribution of the project is the combination of explicit knowledge, inferred knowledge, analytical enrichment and machine-learning representations within one KG. This allows the same graph to support customer analysis, financial analysis, product and seller analysis, sentiment analysis, logical inference, graph embeddings and interactive analytical services.

Overall, the project demonstrates how Knowledge Graph technology can transform interconnected e-commerce data into a reusable structure for analysis, reasoning and downstream intelligent applications.